An electric -bike travels at a top speed of 120 Km/hr, whereas the minimum speed is 30 km/hr. Thus, the average speed at which the electric bike travels is 80 km/hr. If the standard deviation is 9 find the probability of the motorbike with a speed more than 90 km/hr.

Standardized Value (X)	90
Mean (μ)	80
Standard Deviation (σ)	9

Calculate the standard normal distribution using the formula below

Z = (X - μ) / σ	**1.1111**

Calculate the probability using the normal distribution Excel function, NORM.DIST.

Cumulative Probability is a logical function denoting the type of distribution where TRUE outputs the NORMAL DISTRIBUTION function and FALSE outputs the PROBABILITY DENSIY function.

P(X≤x)= NORM.DIST(x, mean, standard_dev,TRUE)	**0.8667**

OR

P(X≤x)= NORM.S.DIST(Z, TRUE)	**0.8667**

Answer

Probability of **less than X ->** =	0.8667
Probability of **greater than X** -> (**1 - p)** =	0.1333
So, probability of electric -bike travelling with a speed higher than 90 KM PER HR is (in %)	13.32602629